

RAHUL SHARMA

AUTOMATION ENGINEER | PLAYWRIGHT · PYTHON · PYTEST · CI/CD

Chandigarh, India | +91 78072 32349 | fastasf48@gmail.com

Portfolio: fastasf48-hash.github.io | github.com/fastasf48-hash | linkedin.com/in/rahul-sharma-65093b1b9

PROFESSIONAL SUMMARY

Automation Engineer with **4+ years** in software quality assurance — one year manual, then three-plus years building test automation frameworks. Specialises in **Playwright with Python and Pytest** across UI, REST API and database layers in one maintainable codebase, wired into **Jenkins and GitHub Actions** CI/CD. Currently leading a legacy Selenium-to-Playwright migration for an enterprise airport access-control product. Framework architecture is the focus — Page Object Model, scoped fixtures, parameterised data, configuration-driven environments. Four public, documented framework repositories: fastasf48-hash.github.io.

CORE SKILLS

Test Automation: Playwright, Pytest (fixtures, markers, parameterization, hooks), Page Object Model (POM), Selenium WebDriver, TestNG, data-driven and keyword-driven testing, cross-browser testing (Chromium, Firefox, WebKit), parallel execution (pytest-xdist)

Languages & API Testing: Python, SQL, Java · REST API automation, Playwright APIRequestContext, Requests, JSON Schema validation (jsonschema), Faker, pytest-playwright, Basic and Bearer token authentication

CI/CD & Tooling: Jenkins, GitHub Actions, Docker, Git, GitHub, JIRA, Allure, pytest-html, screenshot / video / trace capture on failure

Databases: Microsoft SQL Server, MySQL, PostgreSQL, SQLite, SQLAlchemy, parameterized queries, backend data-integrity validation

Platforms & Practices: Alert Enterprise Guardian (PIAM), Apache NiFi, Apache ZooKeeper, Tomcat · SDLC, Agile / Scrum, OOP, Data Structures & Algorithms, test strategy, defect triage

PROFESSIONAL EXPERIENCE

Alert Enterprise — Chandigarh, India

Software Engineer II

Jul 2024 – Present

- Lead the migration of the legacy Selenium regression suite for **Airport Guardian** to a Playwright + Python + Pytest framework, improving execution speed and sharply reducing flaky failures.
- Architected the replacement framework around the **Page Object Model**, scoped Pytest fixtures and parameterised test data, so UI and API suites share one reusable, maintainable spine.
- Built REST API coverage using **Requests, jsonschema and Faker** — request/response assertions, JSON Schema validation, and data-driven cases fed by dynamically generated data.
- Integrated regression and API suites into **Jenkins CI/CD pipelines** for continuous testing on every merge request, and added **SQLAlchemy**-based database checks verifying backend data integrity inside end-to-end scenarios.
- Introduced structured **Allure and HTML reporting** plus centralised logging; mentor QA engineers on Python, Pytest and Playwright fundamentals.

Software Engineer I

Jan 2022 – Jun 2024

- Achieved a **98% defect detection rate** on Airport Guardian through rigorous test design and execution, validating enterprise PIAM workflows built on Apache NiFi, Apache ZooKeeper, Tomcat and Microsoft SQL Server.
- Wrote Selenium (Java) automation for critical regression scenarios over approximately six months, cutting manual effort and test-cycle duration significantly — by **100% in several suites**.
- Partnered directly with customers on QA feedback and ran process training for support teams, reducing customer-reported incidents by **30%**.

SELECTED PROJECTS

Playwright + Pytest Automation Framework github.com/fastasf48-hash/playwright-pytest-automation-framework

- UI, REST API and database suites in one repository: six page objects on a shared **BasePage**, a full CRUD-plus-token-auth lifecycle on Playwright's APIRequestContext, and SQLite validation via parameterized queries. Multi-environment **--env** config, JSON Schema validation, Allure reporting with trace-on-failure, and parallel xdist runs across Chromium, Firefox and WebKit from GitHub Actions, Docker and Jenkins.

AI Test Failure Analyzer for Playwright github.com/fastasf48-hash/AI-Test-Failure-Analyzer-For-Playwright

- Captures screenshot, video, trace, console/network logs and stack trace on every failure, triaged by a **free deterministic classifier**. An opt-in LLM layer returns a **Pydantic-validated** root-cause report — category, confidence, severity, suggested fix, improved locator — behind a single auditable call site, with a Streamlit dashboard over a SQLAlchemy 2.0 schema.

Playwright Accessibility Toolkit github.com/fastasf48-hash/playwright-accessibility-toolkit

- Automated **WCAG 2.x** scanning on axe-core: typed violation objects grouped by severity plus a configurable threshold assertion that fails the build, verified against both a clean and a deliberately inaccessible fixture page.

EDUCATION

B.Tech, Computer Science — Chandigarh Engineering College, Punjab

8.1 CGPA · 2022

Senior Secondary & Secondary (CBSE) — DAV Dharamshala, Himachal Pradesh

Class XII 81.6% · Class X 9.2 CGPA

AWARDS

Quality Champion — awarded as a fresher for a 98% defect detection rate. · **Outstanding Performer** — two consecutive appraisal cycles for delivery on the Airport Guardian automation programme.